Classification

- **Introduction to Classification:**
- Classification is the supervised machine learning task which learns to assign an input to the class [predefined set of categories].

- Spam / Not Spam
- Cat / Dog.

- **Classifier:** trained classification model: spam detector.
- **Binary Classification:** predicting between two possible classes.
  eg :- Spam / Non-Spam
  - Defective / Non-Defective
  - Person / no. person.
  - Fault / Normal.

- **Multi-class classification:** predicting between more than two possible classes
  eg - Normal / Overheating / Electrical Fault.
  - Digit Recognition [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

- Multi-class: choose only one option. [which is main]: person.
- Multi-label: choose available one/more true other options.
  - car, person

- **class label:** It identifies the category associated with a sample.
  eg:
  labels: [0, 1, 0, 1, 0]

- For **logistic regression**, probabilities are calculated by sigmoid functions.

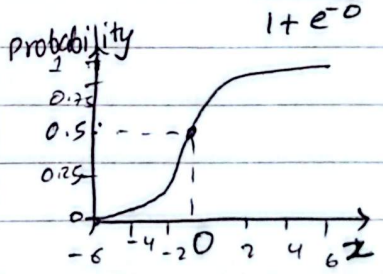
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$\sigma(z)$ → [0-1]

where;
z = score calculated from the input features and learned model parameters.

when, $\sigma(0)$: $z = 0$

$$\sigma(0) = \frac{1}{1 + e^{-0}} = \frac{1}{2} = 0.5.$$



z	-3	-2	-1	0	1	2	3
$\sigma(z)$	0.047	0.119	0.269	0.500	0.731	0.881	0.953

Decision Threshold

probability → final class prediction,
0 = Normal
1 = fault.

i.e 0 → non-spam [0-0.5)
1 → spam [0.5-1].

- What if important mail goes in spam folder, we can ignore it. So;

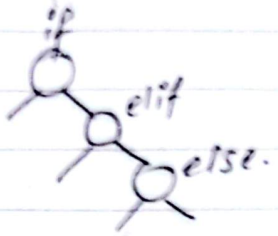
decision threshold = 0.75 (we can change)

so; 0.-0.75 : non-spam
0.75-1 : Spam.

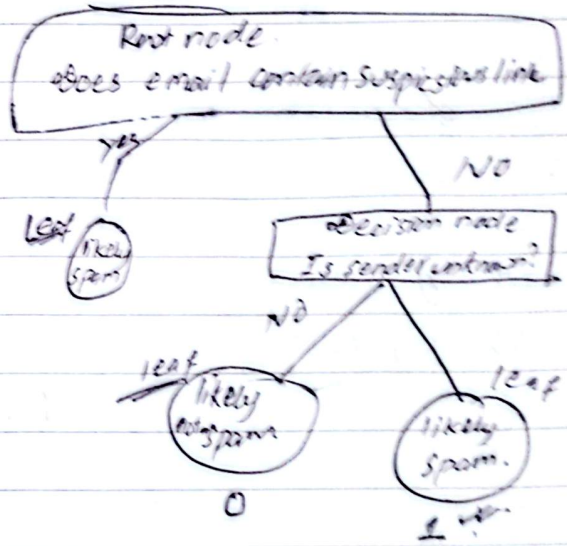
- likewise example: problem with 0.5
  - Medical screening → avoid positive case
  - Industrial fault detection → avoid dangerous fault.
  - security monitoring → ~~don't~~ miss intrusions.

- Five classification Application
  - Industrial defect detection.
  - Smart surveillance.
  - Hardware fault detection.
  - Network intrusion detection.
  - Handwritten digit recognition.

Decision Tree



- It learns a sequence of questions from training data and uses those questions to predict a result.



In ML, we don't write these questions, ML algorithm learns which questions ~~and thresholds~~. Here, thresholds best separate the training examples.

- Solve both:
  - Classification &
  - Regression Problems

How does the algorithm decide which question to ask?

→ It tries possible splits and measures how well each split separates the classes.

a) Gini Impurity:

It measures how mixed the classes are in group of samples.

- Gini = 0 means all samples belong to one class.
- Higher Gini means the group contains a mixture of classes.

$$Gini = 1 - \sum_{i=1}^{C} P_i^2$$

where:

- C = no. of classes.
- P_i = ~~Portion~~ Proportion of samples belonging to class i

Eg 1: Pure group:

10 samples:

- 10 = spam ~~belong to~~
- 0 = not-spam.

$$Gini = 1 - (1^2 + 0^2) = 0$$

$(\frac{10}{10})^2$ $(\frac{0}{10})^2$

- **Eg 2:** Mixed group

10 samples: - 5 = spam
- 5 = non-spam

$$Gini = 1 - (0.5^2 + 0.5^2) = 0.5$$

$(\frac{5}{10})^2$ $(\frac{5}{10})^2$.

b) Entropy:

It is another way to measure how mixed a group is.

$$H = -\sum_{i=1}^{c} P_i \log_2(P_i)$$

Distribution	Entropy
10 A, 0 B	0
8 A, 2 B	0.722
5 A, 5 B	1

c) Information Gain

How does the tree decide whether a split is useful?

Information Gain = H(Parent) - H(children)

It measures reduce in entropy after a split.

- The tree needs to find a splitting point that separate the two classes as well as possible.

- But how does it know which split is best?

Entropy /

- Gini impurity is used because the tree needs to measure how good or bad, a group is after splitting.
- Information gain - How much did our split improve things.

This is for Decision Tree Classifier

- Control Overfitting in Decision Tree?
  - max_depth:
  - min_samples_split: require a minimum number of samples before splitting a node.
  - min_samples_leaf: Require a minimum number of samples in a leaf.